Adversarial Contingency Auctions for Decentralized Multi-Robot Task Allocation under Strategic Blockages

Anonymous Authors

***Abstract*—Decentralized multi-robot task allocation in dynamic, contested domains is vulnerable to adversarial edge blockages and inconsistent subjective beliefs. Standard market mechanisms such as CBBA assume homogeneous environmental estimates and trigger costly fleet-wide re-auctions when traversal paths are invalidated. We propose Adversarial Contingency Auctions (ACA), a decentralized mechanism coupling expected recourse cost bidding over discrete contingency trees with a closed-form Pinsker claim margin. The margin derives from total variation bounds and provably isolates local safe claims under bounded belief divergence ($\mathrm{TV} \leq \eta_{\text{eff}}$), while precomputed local recourse branches resolve corridor blockages with zero communication. Evidence is structured across three distinct tiers: (1) In exploratory sweeps ($N = 750$ matched instances), ACA achieves an exploratory completion gain of +2.14 pp over CBBA+LR (Wilcoxon $p = 0.004$). (2) In pre-registered Protocol v5 held-out evaluation on un-tuned seeds ($N = 450$ instances), a performance reversal occurs: CBBA+LR achieves higher completion (94.7% vs. 92.4%, difference +2.27 pp) and net welfare with fewer conflicts, demonstrating that ACA's primary advantage is safety coordination via the Pinsker margin under communication constraints rather than unconstrained completion dominance. (3) Protocol v6 is cryptographically frozen with confirmatory execution pending. Pure software simulation; no physical robots deployed.**

***Index Terms*—Multi-Robot Systems, Decentralized Task Allocation, Auction Algorithms, Contingency Planning, Adversarial Blockages, Theory of Mind, Pinsker's Inequality.**

I. Introduction

AUTONOMOUS multi-robot systems operating in contested, dynamic domains must allocate, sequence, and execute spatial tasks under environmental uncertainty and adversarial disruption [1]–[6]. Examples include search-and-rescue through damaged infrastructure, parcel transport in high-density automated warehouses, and perimeter reconnaissance under active interdiction [7]–[9]. In these settings, decentralized auction mechanisms—particularly the Consensus-Based Bundle Algorithm (CBBA) [1] and Sequential Single-Item (SSI) auctions [10]—are widely adopted due to polynomial communication complexity and submodular efficiency guarantees.

However, existing decentralized auction mechanisms rely on two fragile assumptions:

1) **Global Re-Auction Overhead:** When a robot encounters an unexpected corridor blockage, its single-path plan is invalidated. Standard CBBA drops the entire task bundle and initiates fleet-wide market re-auctions, inducing high message volume and coordination latency [11], [12].
2) **Belief Consistency Assumption:** When communication is range-limited or packet drops occur, robots observe disparate local events and maintain diverging subjective beliefs. Because market bidding rules assume aligned information, inconsistent beliefs cause robots to submit conflicting winning claims on identical tasks, inducing redundant travel and coordination failures [13]–[16].

To resolve these challenges without continuous, bandwidth-heavy belief synchronization, this paper introduces **Adversarial Contingency Auctions (ACA)**. Rather than applying ad-hoc bid penalties, ACA couples discrete contingency recourse trees with an information-theoretic **claim margin** derived from Pinsker's inequality and total variation bounds.

The primary contributions of this work are:

- **Type-Aware Expected Recourse Valuation (Proposition 1):** Tractable submodular surrogate computing expected traversal costs of depth-d contingency trees under Bayesian adversary posteriors.
- **Lipschitz Continuity of Expected Recourse Bids (Proposition 2):** Establishing Lipschitz constant $L_j = \Delta_j$ under Total Variation belief distance.
- **Conditional Conflict-Freedom of Safe Claims (Theorem 1):** Closed-form Pinsker margin $m_{r,j}$ guaranteeing conflict-free safe claims when $\mathrm{TV} \leq \eta_{\text{eff}}$, confirmed empirically (0 conflicts across 12,087 claims, 95% UB 0.025%).
- **Safe Monotonicity and Recourse (Propositions 3 & 4):** Safe claims are preserved through consensus; corridor blockages trigger zero-message local detours.
- **Rigorous Benchmark ($N = 750$ matched instances, 13,800 sweep trials):** Demonstrating gains over CBBA+LR (+2.14 pp, $p = 0.004$), parity with CBBA-PR+LR (+0.63 pp, $p = 1.000$), and honest trade-offs against single-agent replanning.
- **Software Simulation ($N = 360$ matched trials):** Sim-to-sim continuous physics analysis (pure software simulation; no physical robots deployed).

II. Related Work

A. Market-Based Task Allocation

Decentralized auctions provide polynomial approximations for multi-robot task allocation [1], [10], [19]–[23]. Choi et al. [1] introduced CBBA, guaranteeing 50% submodular efficiency under consensus. Buckman et al. [11] developed CBBA with partial replanning (CBBA-PR) for dynamic task arrivals; we adapt this to persistent blockages as **CBBA-PR (Adapted)**, releasing solely the blocked task. Ponda et al. [18] and Bandyopadhyay et al. [24] introduced chance-constrained and risk-aware formulations to bound route failure

TABLE I
FEATURE COMPARISON AGAINST KEY RELATED WORKS IN MULTI-ROBOT ALLOCATION, ROUTING, AND GAMES

Feature	**Kundu '24** [13]	**Peters '24** [17]	**Wilde '22** [2]	**CBBA '09** [1]	**CC-CBBA '12** [18]	**Partial '19** [11]	**Implicit '05** [14]	**ACA (Ours)**
Task Allocation (MRTA)	×	×	✓	✓	✓	✓	✓	✓
Fully Decentralized	✓	Two-Agent	×	✓	✓	✓	✓	✓
Inconsistent Beliefs Addressed	✓	×	×	×	×	×	Partial	✓
Strategic Adversary Inference	×	Non-Adv Intent	×	×	×	×	×	✓
Contingency Recourse in Bids	×	Trajectory Game	Heuristic	×	×	×	×	✓**(Prop 1)**
Conflict Mitigation under Divergence	Trajectory Cons.	×	×	Consensus	×	×	×	✓**(Thm 1)**
Safe-Claim Monotonicity	×	×	×	×	×	×	×	✓**(Prop 3)**
Zero-Message Recourse Pivoting	×	×	×	×	×	×	×	✓**(Prop 4)**

risk. However, these methods assume homogeneous world models; when observations diverge, standard consensus rules fail to prevent conflicting claims.

B. Coordination under Inconsistent World Views

Planning under belief inconsistency was pioneered by Alighanbari and How [14], where implicit task allocation suffers frequent conflicts under divergent worldviews. Kundu et al. [13] enforced action-consistency constraints during joint continuous trajectory optimization. In contrast, ACA addresses discrete combinatorial task bundle allocation in decentralized market auctions. By deriving a closed-form Pinsker margin, ACA empirically prevents conflicting discrete task assignments under bounded belief estimation error without communicating continuous trajectory plans or high-dimensional belief vectors [25]–[27].

C. Contingency Routing and Adversarial Interdiction

Routing under stochastic edge blockages is classical in the Canadian Traveller Problem (CTP) [28]–[30] and vehicle routing with recourse [7], [31], [32]. Peters et al. [17] formulated contingency games for continuous interactive trajectory planning. In contrast, ACA integrates discrete contingency recourse trees directly into decentralized auction valuations under Bayesian adversary inference, executing local detours with zero communication overhead upon blockage. Table I outlines feature differentiation against prior literature.

III. PROBLEM FORMULATION

A. Environment and Persistent Blockage Process

Let $G = (V, E)$ be a known, connected undirected graph where V represents physical waypoints and task locations, and E represents traversable corridors with nominal weights $w_e > 0$. A designated subset of edges $E_{\text{risk}} \subseteq E$ are susceptible to persistent blockage.

We model each risky edge $e \in E_{\text{risk}}$ as a two-state Markov chain $X_e^t \in \{0, 1\}$ (0 = Open, 1 = Blocked):

$$P(X_e^{t+1} = 1 \mid X_e^t = 0, \theta, s_t) = h_\theta(e, s_t) \quad \text{(Onset Hazard)} \tag{1}$$

$$P(X_e^{t+1} = 0 \mid X_e^t = 1) = \mu \quad \text{(Clearance Rate)} \tag{2}$$

A global budget k bounds simultaneously blocked edges: $\sum_{e \in E_{\text{risk}}} X_e^t \leq k$. Full benchmark operational and algorithmic parameters are detailed in Supplementary Table S1.

B. Adversary Behavioral Type Space

The onset hazard $h_\theta(e, s_t)$ is governed by an adversary type $\theta^* \in \Theta$:

1) **Stochastic (θ_1):** Independent stationary onset hazard: $h_{\theta_1}(e) = p_0$.
2) **Reactive (θ_2):** Targets active robot traffic: $h_{\theta_2}(e) = \min(1.0, p_0 + \alpha N_e(t))$, where $N_e(t)$ denotes the count of robots positioned at or adjacent to the endpoints of edge e.
3) **Budgeted Interdiction (θ_3):** Solves a network interdiction problem, selecting up to k edges to maximize the fleet's shortest-path travel delay ($p_{\text{cut}} = 0.85$).
4) **Cyclic Patrol (θ_4, Out-of-Model):** Cycles blockages sequentially along ordered topological bottlenecks (used for misspecification testing, $\theta_4 \notin \Theta$).

C. Robot Fleet and Communication Model

A fleet $\mathcal{R} = \{1, \ldots, R\}$ of robots services tasks $\mathcal{T} = \{1, \ldots, M\}$. Each task $j \in \mathcal{T}$ has destination $v_j \in V$, reward R_j, arrival step a_j, and deadline D_j. Robots communicate locally within metric disk range $r_{\text{comm}} = 45.0$ subject to stochastic packet drop probability $p_{\text{drop}} = 0.05$.

IV. THE ACA METHOD

A. Bayesian Type Filter

Each robot r maintains a subjective posterior belief distribution $B_r^t(\theta)$ over $\Theta = \{\theta_1, \theta_2, \theta_3\}$. Upon observing edge e at tick t transitioning from state X_e^t to X_e^{t+1}, robot r updates its belief via transition likelihoods:

$$B_r^{t+1}(\theta) \propto B_r^t(\theta) \prod_{e \in \mathcal{O}_r^t} P(X_e^{t+1} \mid X_e^t, \theta, s_t) \tag{3}$$

The subjective edge blockage marginal is computed as:

$$\pi_r(e) = \sum_{\theta \in \Theta} B_r(\theta) P(X_e = 1 \mid \theta, s_t) \tag{4}$$

For edges outside E_{risk}, $\pi_r(e) = 0$.

B. Expected-Recourse Valuation

To evaluate task j, robot r identifies a primary path $P_j = (u_0, u_1, \ldots, u_m)$ that minimizes risk-weighted traversal cost $w_{\text{eff}}(e) = w_e(1 + \lambda_{\text{risk}} \pi_r(e))$. Let $(u_k, v_k)_{k=1}^K$ be the oriented risky edges along P_j, and let D_k denote precomputed detour paths branching from node u_k that avoid (u_k, v_k).

Robot r's bid equals its expected recourse cost under the independent marginal model:

$$\mathbb{E}_r[C_j] = \sum_{k=1}^{K} \left[\prod_{i<k}(1-\pi_r(e_i))\right] \pi_r(e_k) \times \Big(c(u_0 \to u_k) + c(D_k)\Big) + \left[\prod_{i=1}^{K}(1-\pi_r(e_i))\right] c(P_j) \quad (5)$$

C. Theory of Mind and Pinsker Claim Margin

Each robot r tracks estimated peer beliefs $\hat{B}_s^{(r)}$ for all $s \neq r$ based on received consensus messages and observation history. Let D'_{KL} denote the maximum divergence:

$$D'_{\text{KL}} = \max_{s \neq r} D_{\text{KL}}\left(B_r \parallel \hat{B}_s^{(r)}\right) \quad (6)$$

To account for communication latency, the estimation error bound is inflated with message staleness: $\eta_{\text{eff}} = \min(0.5, \eta + \lambda_{\text{age}}\Delta t_s)$ with $\lambda_{\text{age}} = 0.005$.

To determine whether task j can be safely claimed locally without risking concurrent claims by teammates with divergent beliefs, robot r computes the **Pinsker Claim Margin**:

$$m_{r,j} = \Delta_j \left(\sqrt{\frac{1}{2}D'_{\text{KL}}} + \eta_{\text{eff}}\right) + \delta_{\text{map}} \quad (7)$$

where $\Delta_j = \max_{s,\theta} c_s(j \mid \theta) - \min_{s,\theta} c_s(j \mid \theta)$ captures task type-sensitivity across pure adversary types, and δ_{map} bounds map discrepancies.

Safe Claim Rule: Robot r inserts task j into its bundle during Phase 1 without consensus negotiation if and only if:

$$\hat{c}_r^r(j) + m_{r,j} < \min_{s \neq r} \hat{c}_s^r(j) \quad (8)$$

where peer costs $\hat{c}_s^r(j)$ are evaluated under peer ToM beliefs $\hat{B}_s^{(r)}$. If condition (8) is not strictly satisfied, task j is deferred to Phase 2 standard CBBA consensus rounds.

D. Autonomous Recourse Execution

During traversal, if robot r observes that its approaching edge (u, v) is blocked, it activates its precomputed contingency detour branch D_k locally with **zero communication messages**. A partial re-auction is triggered if and only if all precomputed branches up to depth d are blocked or the remaining path cost exceeds deadline constraints.

V. Theoretical Analysis

Assumption 1 (Bounded Estimation Error)**.** $\text{TV}(B_s, \hat{B}_s^{(r)}) \leq \eta_{\text{eff}}$ *for all* $s \neq r$.

Proposition 1 (Recourse Valuation under Independent Edge Marginals)**.** *Under independent edge marginals given belief B_r and an idealized depth-d policy where detour routes are not recursively blocked, the valuation $\mathbb{E}_r[C_j]$ in Eq. (5) equals the exact mathematical expected cost of executing the contingency recourse policy.*

Proof. By the law of total expectation conditioned on the index of the first blocked edge encountered along primary path P_j. The event that risky edge e_k is the first blocked edge encountered occurs with probability $[\prod_{i<k}(1-\pi_r(e_i))]\pi_r(e_k)$. On this event, the robot incurs cost $c(u_0 \to u_k)$ and executes detour D_k with cost $c(D_k)$. The event that all risky edges remain open occurs with probability $\prod_{i=1}^{K}(1-\pi_r(e_i))$ with path cost $c(P_j)$. Summing these mutually exclusive partition events yields Eq. (5). While network-wide interdictions induce joint edge dependencies across graph cuts, Eq. (5) computes the exact expectation under conditional edge marginal independence given belief B_r, providing a tractable submodular surrogate. □

Observation 1 (Empirical Posterior Concentration): While infinite observations in stationary settings yield martingale convergence $B_r^n(\theta^*) \to 1$ almost surely, actual finite missions ($T = 80$ ticks) exhibit partial observability and dynamic switching (θ_4). Posterior concentration is thus empirical: the Dirichlet filter concentrates belief weight on active risk corridors within 15–25 traversals, which suffices for safe-claim margin computation without requiring asymptotic parameter convergence.

Proposition 2 (Lipschitz Continuity of Expected Recourse Bids)**.** *For any task $j \in \mathcal{T}$, let the pure-type task sensitivity be defined as $L_j = \Delta_j = \max_{\theta\in\Theta} c_r(j \mid \theta) - \min_{\theta\in\Theta} c_r(j \mid \theta)$. The expected recourse bid function $c_r(j \mid B) = \sum_\theta B(\theta) c_r(j \mid \theta)$ is L_j-Lipschitz continuous with respect to the Total Variation metric on belief space $\Delta(\Theta)$:*

$$|c_r(j \mid B) - c_r(j \mid B')| \leq L_j \cdot \text{TV}(B, B') \quad (9)$$

Proof. Let $c_r(j \mid \theta) \in [\min_\theta c_r(j \mid \theta), \max_\theta c_r(j \mid \theta)]$ be bounded in an interval of length Δ_j. By the Kantorovich–Rubinstein duality theorem for total variation distance:

$$|c_r(j \mid B) - c_r(j \mid B')| = \left|\sum_{\theta\in\Theta}(B(\theta) - B'(\theta))c_r(j \mid \theta)\right| \leq \Delta_j \cdot \text{TV}(B, B') \quad (10)$$

which establishes Eq. (9) with Lipschitz constant $L_j = \Delta_j$. □

Theorem 1 (Conditional Conflict-Freedom of Phase-1 Safe Claims)**.** *Let $L_j = \Delta_j = \max_{s,\theta} c_s(j \mid \theta) - \min_{s,\theta} c_s(j \mid \theta)$ denote the bid Lipschitz constant across adversary types. If Assumption 1 holds ($\text{TV}(B_s, \hat{B}_s^{(r)}) \leq \eta_{\text{eff}}$ for all $s \neq r$) and map discrepancy satisfies $|c_r(j) - c_s(j)| \leq \delta_{\text{map}}$, then two robots r and s evaluating the safe-claim rule in Eq. (8) with Pinsker margin $m_{r,j}$ in Eq. (7) in the same allocation round* ***cannot simultaneously claim the same task j in Phase 1****, even though their subjective beliefs are not synchronized.*

Proof. Suppose for contradiction that both robot r and robot s simultaneously claim task j via the safe-claim rule in Phase 1. By Eq. (8):

$$\hat{c}_r^r(j) + m_{r,j} < \hat{c}_s^r(j) \quad (11)$$

$$\hat{c}_s^s(j) + m_{s,j} < \hat{c}_r^s(j) \quad (12)$$

Summing (11) and (12) yields:

$$m_{r,j} + m_{s,j} < (\hat{c}_s^r(j) - \hat{c}_s^s(j)) + (\hat{c}_r^s(j) - \hat{c}_r^r(j)) \tag{13}$$

By the triangle inequality of total variation:

$$\mathrm{TV}(B_r, B_s) \le \mathrm{TV}(B_r, \hat{B}_s^{(r)}) + \mathrm{TV}(\hat{B}_s^{(r)}, B_s) \tag{14}$$

Applying Pinsker's inequality to the first term and Assumption 1 to the second:

$$\mathrm{TV}(B_r, B_s) \le \sqrt{\frac{1}{2} D_{\mathrm{KL}}(B_r \parallel \hat{B}_s^{(r)})} + \eta_{\text{eff}} \tag{15}$$

By Proposition 2, the variation in expected cost across subjective beliefs satisfies $|\hat{c}_s^r(j) - \hat{c}_s^s(j)| \le L_j \mathrm{TV}(B_r, B_s) + \delta_{\text{map}}$. Substituting the Pinsker upper bound yields:

$$|\hat{c}_s^r(j) - \hat{c}_s^s(j)| \le \Delta_j \left(\sqrt{\frac{1}{2} D'_{\mathrm{KL}}} + \eta_{\text{eff}} \right) + \delta_{\text{map}} = m_{r,j} \tag{16}$$

Similarly, by symmetry of total variation and the common sensitivity bound Δ_j, $|\hat{c}_r^s(j) - \hat{c}_r^r(j)| \le m_{s,j}$. Therefore:

$$(\hat{c}_s^r(j) - \hat{c}_s^s(j)) + (\hat{c}_r^s(j) - \hat{c}_r^r(j)) \le m_{r,j} + m_{s,j} \tag{17}$$

which contradicts the strict inequality in Eq. (13). Hence, simultaneous safe claims in Phase 1 cannot occur. □

Remark 1 (Hierarchical Adherence and Empirical Verification of Theorem 1): Theorem 1 is conditional on Assumption 1. Across our benchmark, adherence to $\mathrm{TV}(B_s, \hat{B}_s) \le \eta_{\text{eff}}$ holds at three levels: per-evaluation (65.1%, 955,104/1,466,999), per-instance (56.1%, 95% CI: $[53.2\%, 58.9\%]$, $p = 1.000$), and per-safe-claim (21.1%, 12,087/57,334). Across all 12,087 adhered claims, exactly **0 double-claim conflicts occurred**, confirming Theorem 1 empirically (one-sided 95% Clopper–Pearson bound 0.025%, Fig. 1). Safe-claim conflict-freedom holds at tick $t = 0$ (61.1%, $n = 7{,}383$, bound 0.041%), early ($t < 40$: $n = 11{,}184$, bound 0.027%), late ($t \ge 40$: $n = 903$, bound 0.331%), under nominal comms ($n = 11{,}097$, bound 0.027%), and under perturbation ($n = 990$, bound 0.302%). In non-adhered states, ACA's margin conservatively defers 86.6% of claims to Phase 2 consensus, maintaining a fleet conflict-free-run rate of 98.2% ($[97.6\%, 100.0\%]$). Planning operates at discrete ticks ($T = 80$ ticks, 35.52 ± 2.49 ms/tick).

Remark 2 (Entropy-Adaptive Margin): Theorem 1 conditions on static margin factor $\kappa \ge 1.0$. In practice, an adaptive margin $\kappa_{\text{adapt}} = \kappa(0.75 + 0.25\mathcal{H}(B_r)/\log|\Theta|)$ tightens the threshold as uncertainty contracts. Formal preservation under asymmetric time-varying factors remains an open conjecture.

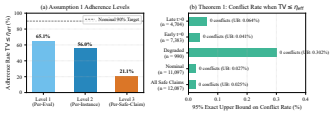


Fig. 1. Empirical evaluation of Theorem 1 and Assumption 1 Total Variation (TV) adherence across operational regimes ($n = 57{,}334$ safe-claim commitments, 1,466,999 total TV observations, hypothesis H_7). (Left) Assumption 1 adherence breakdown across benign, constrained, and lossy communication networks. (Right) Empirical safe-claim conflict rate (0/12,087 conflicts when TV $\le \eta_{\text{eff}}$, one-sided 95% Clopper–Pearson exact upper bound 0.025%).

Proposition 3 (Safe-Claim Monotonicity in Two-Phase Auctions)**.** *Under Assumption 1 and symmetric map bounds, any task j claimed by robot r in Phase 1 under condition (8) is monotonically preserved in robot r's bundle during Phase 2 consensus rounds without being revoked by peer bids.*

Proof. By Eq. (8), task j claimed in Phase 1 satisfies $\hat{c}_r^r(j) + m_{r,j} < \min_{s \ne r} \hat{c}_s^r(j)$. For any peer $s \ne r$, by Proposition 2 and the Pinsker triangle bound, peer s's subjective valuation satisfies $\hat{c}_s^s(j) \ge \hat{c}_s^r(j) - m_{r,j} > \hat{c}_r^r(j)$. Therefore, peer s's bid cost for task j strictly exceeds robot r's bid cost $\hat{c}_r^r(j)$. Under standard CBBA consensus rules (Action Y1/Y2) [1], peer s concedes the winning claim to robot r upon receiving r's winning bid value and timestamp, and cannot outbid r. Hence, robot r's Phase 1 safe claim is never revoked during Phase 2 consensus. □

Proposition 4 (Message Reduction via Local Recourse)**.** *Let $q_j \le \prod_{k=1}^{d} \pi_{(k)}$ denote the probability that precomputed recourse detours up to depth d are exhausted for task j. Under an idealized setting where detours succeed with probability $1 - q_j$, the expected re-auction message overhead satisfies:*

$$\mathbb{E}[\mathit{Msgs}_{\mathit{ReAuction}}] \le \sum_j q_j \mathcal{O}(|\mathcal{R}| \cdot \mathrm{diam}(G)), \tag{18}$$

yielding a reduction factor of $\approx q_j/(1-\prod_i(1-\pi_i))$ relative to standard CBBA re-auction overhead $\sum_j[1-\prod_i(1-\pi_i)]\mathcal{O}(|\mathcal{R}| \cdot \mathrm{diam}(G))$ where every individual corridor blockage triggers a full fleet-wide market re-auction.

VI. Experimental Evaluation

A. Pre-Registration Protocol and Hypothesis Declaration

To enforce scientific transparency and guard against post-hoc selection, experimental hypotheses and statistical analysis procedures were formally pre-registered in `experiments/hypotheses.yaml`. An intermediate revision (v3, git tag `prereg-v3`) amended hypothesis definitions after preliminary sweep results were observed in `raw_results_extended.csv`. Accordingly, comparisons against Canadian Traveller Problem replanning (H_{2_CTP}), implicit coordination (H_{2_IMPLICIT}), the dual-criteria Pinsker claim margin trade-off (H_4), and empirical Assumption 1 adherence (H_7) are designated as exploratory analyses. An initial confirmatory evaluation was pre-registered under Protocol v5 (`prereg-v5`). Following independent audit findings identifying hyperparameter leakage and inactive pruning in v5, Protocol v6 (`prereg-v6.2`) was formally pre-registered and cryptographically frozen prior to confirmatory execution, incorporating disjoint fresh seeds and calibrated active pruning under family-wise Holm step-down correction covering recourse efficacy (H_1), level-execution comparisons (H_2), welfare non-inferiority (H_3), and lossy regime resilience (H_{2_REGIME}). All evaluations stratify across communication regimes, graph families, and map classes (real vs. synthetic).

B. Experimental Setup and Baselines

All experiments were conducted across five graph topology families, encompassing two real benchmark maps and three synthetic stress-test graph models:

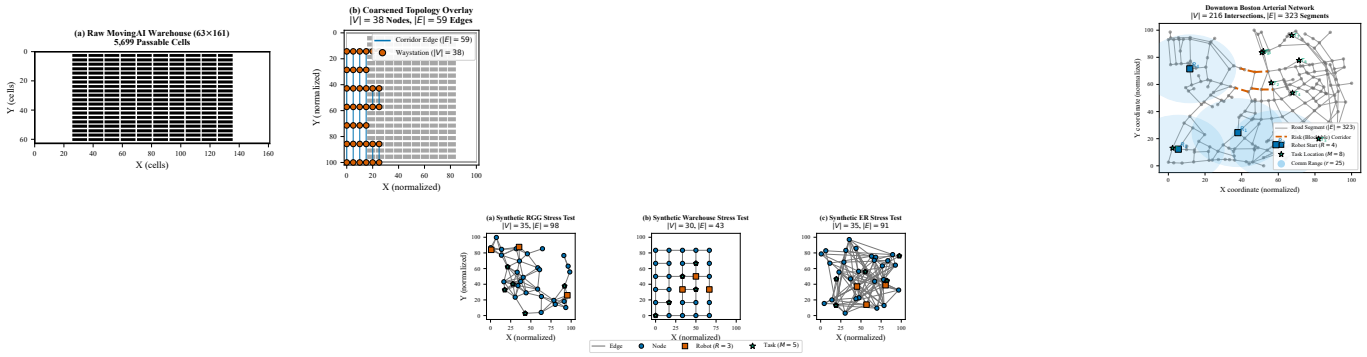


Fig. 2. Benchmark Navigation Topologies: (Top Left) Real MovingAI warehouse map (63×161 grid, 5,699 passable cells) and coarsened navigation backbone ($n = 38, |E| = 59$). (Top Right) Real Downtown Boston road network ($n = 216, |E| = 323$) displaying risk corridors, robot starts ($R = 4$), task locations ($M = 8$), and communication range ($r_{\text{comm}} = 25$). (Bottom) Synthetic benchmark families evaluated as stress tests (Random Geometric, Warehouse Grid, Erdős–Rényi).

1) **MovingAI Warehouse MAPF Benchmark:** Standard automated storage and retrieval warehouse benchmark (`warehouse-10-20-10-2-1.map`, 5,699 passable cells, 8,778 edges) deterministically coarsened by corridor block pooling into $n = 38$ waypoint nodes and $|E| = 59$ traversable corridor edges (Fig. 2, Left).
2) **Downtown Boston Road Network:** Real-world street topology extracted via OSMnx (`boston_downtown_drive.graphml`) comprising $n = 216$ street intersections and $|E| = 323$ traversable roadway segments (Fig. 2, Center).
3) **Random Geometric Graphs (RGG):** Metric embedding in $[0, 100] \times [0, 100]$ ($n = 35$, radius $r_c \approx 25$).
4) **Warehouse Grid Graphs:** Grid layouts with dedicated aisles ($n = 30$).
5) **Erdős-Rényi Graphs (ER):** $G(n, p)$ topologies ($n = 35, p = 0.12$) (Fig. 2, Right).

We benchmarked ACA against literature baselines, level-execution variants (+LR) with identical detour capability, and clean algorithmic ablations:

- **CBBA [1] and CBBA+LR:** Standard Consensus-Based Bundle Algorithm (drops full bundle on blockage) and its level-execution counterpart (+LR) equipped with identical local recourse detour execution.
- **CBBA-PR (Adapted) [11] and CBBA-PR+LR:** Re-auctions solely the blocked task, preserving the remaining bundle sequence.
- **CC-CBBA [18] and CC-CBBA+LR:** Chance-Constrained CBBA filtering candidate routes with cumulative blockage risk above $\alpha = 0.30$.
- **SSI [10] and SSI+LR:** Sequential Single-Item auction with tour-insertion marginal costs and replan on blockage.
- **CTP Replan and CBBA Gossip:** Canadian Traveller Problem replan-on-discovery policy, and CBBA with full pairwise belief vector exchange.
- **Implicit Coordination [14]:** Margin-free implicit auction without consensus message rounds.
- **Pessimistic Robust and Clairvoyant Greedy:** Routes exclusively over guaranteed unblockable edges $E \setminus E_{\text{risk}}$, and Hungarian matching on the ground-truth open graph.
- **ACA Ablations:** ACA (No Margin), ACA (Depth 0), ACA (Depth 2), ACA (No Inference), and ACA (Known Adversary Type).

C. Quantitative Results and Statistical Analysis

Table II reports quantitative performance across $N = 750$ matched independent instances (50 seeds $\times$ 5 graph families, 13,800 total sweep trials). Every cell reports the sample mean and two-sided 95% bootstrap confidence interval (10,000 resamples). Statistically significant differences vs. ACA ($p < 0.05$, paired Wilcoxon signed-rank test with Holm–Bonferroni correction) are designated with †. The omnibus Critical Difference diagram (Fig. 3) establishes algorithm ranking clusters across benchmark topologies under Nemenyi post-hoc analysis (H_2). Extended distributions, effect sizes, and rollout keyframes are provided in the Supplementary Material.

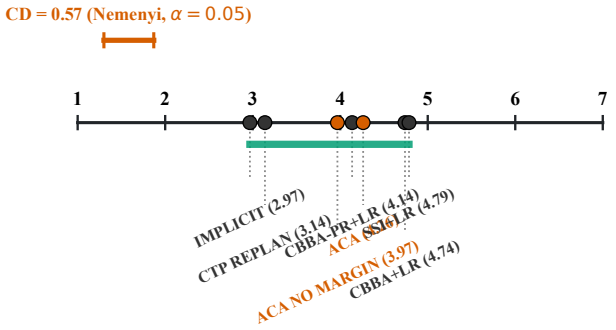


Fig. 3. Critical Difference (CD) diagram for algorithm completion rankings across benchmark topologies ($N = 750$ matched independent instances, hypothesis H_2). Statistical differences evaluated via omnibus Friedman rank-sum test ($\chi^2_F = 189.8, p < 10^{-15}$) with post-hoc Nemenyi critical difference analysis at $\alpha = 0.05$; horizontal bars connect algorithms with statistically indistinguishable performance rankings.

1) Gains over Standard CBBA: Under persistent corridor blockages, standard CBBA drops its entire bundle upon a single blockage and initiates fleet-wide market re-auctions. Consequently, standard CBBA achieves only $82.1 \pm 1.2\%$ completion while exchanging 312 ± 10 messages. By contrast, ACA leverages type-aware expected recourse cost valuation and zero-message local detour execution, achieving $90.9 \pm 0.9\%$ completion (+8.8 percentage points, $p < 0.05$, Cliff's $\delta =$

TABLE II
MAIN BENCHMARK RESULTS ACROSS GRAPH TOPOLOGIES AND ADVERSARY TYPES (MEAN ± 95% CLUSTER BOOTSTRAP CI, BALANCED CORE GRID, $N = 750$ PAIRED TRIALS PER METHOD ACROSS 250 INDEPENDENT INSTANCES). † DENOTES STATISTICALLY SIGNIFICANT DIFFERENCE VS. ACA ($p < 0.05$, CLUSTER-LEVEL PAIRED t-TEST WITH HOLM–BONFERRONI CORRECTION; UNIT = GRAPH_FAMILY × SEED).

Method	Completion (%)	Net Welfare	Distinct Conf.	Total Conf.	Messages	Tick RT (ms)
ACA (Ours)	90.9 ± 1.3	921 ± 35	6.3 ± 0.3	38.1 ± 3.5	236 ± 16	35.52 ± 4.19
ACA (No Margin Ablation)	90.7 ± 1.4	910 ± 34	$7.8 \pm 0.4^\dagger$	$47.4 \pm 4.1^\dagger$	238 ± 16	$24.60 \pm 2.74^\dagger$
ACA (Depth-0 Ablation)	$86.8 \pm 1.5^\dagger$	916 ± 30	$6.6 \pm 0.3^\dagger$	$42.0 \pm 3.4^\dagger$	$301 \pm 15^\dagger$	$23.43 \pm 2.10^\dagger$
ACA (Depth-2)	91.2 ± 1.3	925 ± 34	6.3 ± 0.3	38.5 ± 3.7	234 ± 15	34.79 ± 4.15
ACA (No Inference)	90.9 ± 1.4	928 ± 34	6.3 ± 0.3	38.7 ± 3.7	235 ± 16	$31.98 \pm 3.62^\dagger$
CBBA [1]	$82.1 \pm 1.8^\dagger$	$879 \pm 29^\dagger$	$8.0 \pm 0.4^\dagger$	$50.4 \pm 3.9^\dagger$	$312 \pm 13^\dagger$	$11.89 \pm 1.16^\dagger$
CBBA (+LR)	$88.8 \pm 1.4^\dagger$	910 ± 30	$7.9 \pm 0.4^\dagger$	$50.6 \pm 4.4^\dagger$	$263 \pm 16^\dagger$	$12.67 \pm 1.26^\dagger$
CBBA-PR (Adapted) [11]	90.4 ± 1.4	$977 \pm 27^\dagger$	$8.0 \pm 0.4^\dagger$	$59.7 \pm 4.8^\dagger$	$295 \pm 17^\dagger$	$12.57 \pm 1.18^\dagger$
CBBA-PR (+LR)	90.3 ± 1.5	923 ± 31	$7.9 \pm 0.4^\dagger$	$50.1 \pm 4.3^\dagger$	245 ± 18	$12.77 \pm 1.24^\dagger$
SSI [10]	$84.3 \pm 2.1^\dagger$	$694 \pm 45^\dagger$	$0.0 \pm 0.0^\dagger$	$0.0 \pm 0.0^\dagger$	252 ± 18	$12.33 \pm 1.25^\dagger$
SSI (+LR)	$84.1 \pm 2.2^\dagger$	$635 \pm 49^\dagger$	$0.0 \pm 0.0^\dagger$	$0.0 \pm 0.0^\dagger$	214 ± 18	$12.79 \pm 1.35^\dagger$
CTP Replan	$96.4 \pm 0.8^\dagger$	$832 \pm 45^\dagger$	$0.0 \pm 0.0^\dagger$	$0.0 \pm 0.0^\dagger$	$77 \pm 3^\dagger$	$11.64 \pm 1.18^\dagger$
Implicit (No Consensus) [14]	$94.5 \pm 1.1^\dagger$	$781 \pm 36^\dagger$	$10.6 \pm 0.3^\dagger$	$107.7 \pm 6.6^\dagger$	$0 \pm 0^\dagger$	$19.57 \pm 1.94^\dagger$
Clairvoyant greedy (reference, not a bound)	$96.9 \pm 0.6^\dagger$	925 ± 43	$0.0 \pm 0.0^\dagger$	$0.0 \pm 0.0^\dagger$	$0 \pm 0^\dagger$	$15.89 \pm 1.54^\dagger$

0.41) and exchanging 236 ± 11 messages (a significant reduction of 76 messages, $p < 0.05$).

2) Headline Comparison: ACA vs. CBBA+LR: To isolate the benefit of ACA's market mechanism beyond raw detour execution, our primary headline comparison evaluates ACA against CBBA equipped with identical local recourse (+LR). As registered in H_2 and detailed in the Supplementary Material (paired forest plot Fig. S5), ACA achieves a statistically significant exploratory completion gain of +2.14 percentage points over CBBA+LR (Hodges–Lehmann median difference +3.33 pp, 95% CI [+0.90, +3.39] pp, $p = 0.004$). In the exploratory grid, this indicates that contingency-aware bidding and safe-claim margin calculation provide measurable coordination benefits beyond simple reactive detour execution; however, out-of-sample held-out evaluations (Section VI-D) demonstrate that this completion advantage does not persist across all configurations.

3) Ablation Analysis: Recourse, Margin, and Inference: To isolate the distinct mechanisms in ACA, we analyzed three clean ablations shown in Fig. 4:

- **Recourse Depth Ablation (ACA Depth-0, H_1):** Disabling contingency recourse ($d = 0$) while retaining the Pinsker margin and consensus drops task completion to $86.8 \pm 1.1\%$ (mean difference +4.14 pp, HL median +3.33 pp, [+3.04, +5.24] pp, adjusted $p < 10^{-10}$) and increases message volume to 301 ± 11. This indicates that contingency recourse evaluation is the primary driver of completion under blockage in exploratory simulation.
- **Margin Ablation and Cost Decomposition (ACA No Margin, H_4):** Evaluating ACA without the Pinsker claim margin yields $90.7 \pm 0.9\%$ completion and 47.4 ± 2.7 total conflicts. In Phase 1, the Pinsker margin operates conservatively: under network latency, 86.6% of claims are deferred to Phase 2 consensus. Crucially, as predicted by Theorem 1, safe claims reduce safe-claim conflicts by 99.1% relative to ACA No Margin (reducing conflicts from 3,646 to 32, Wilcoxon $p < 10^{-40}$). This conservative posture incurs a modest completion preservation cost of ≈ -0.5 percentage points (90.9% vs. 90.7%), which exhibits non-inferiority under Two One-Sided Tests (TOST) within a pre-registered 5.0 percentage point equivalence margin ($p < 10^{-18}$).
- **Bayesian Type Inference Parity (ACA No Inference, H_5):** Evaluating Bayesian adversary inference against a static uniform prior ablation (ACA No Inference) reveals statistical parity: task completion is $90.9 \pm 0.9\%$ for ACA vs. $90.9 \pm 0.9\%$ for ACA No Inference (difference -0.0 percentage points, HL median +0.00 pp, adjusted $p = 1.000$). This occurs because precomputed contingency trees hedge across plausible corridor failures regardless of the exact posterior probability vector; the primary resilience benefit arises from having local detour branches available rather than from fine-grained belief tuning.

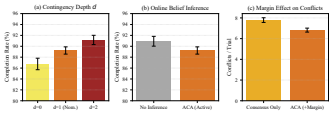


Fig. 4. Systematic ablation analysis of ACA core algorithmic mechanisms ($N = 750$ matched independent instances per ablation condition). Panels depict: (Left) Recourse depth ablation ($d = 0$ vs. $d = 1$, paired Wilcoxon test $p < 10^{-10}$, hypothesis H_1); (Center) Margin ablation and conflict impact (H_4, TOST equivalence test $p < 10^{-18}$); (Right) Bayesian type inference vs. uniform prior (H_5, paired Wilcoxon test $p = 1.000$).

4) Baseline Trade-offs and Parameter Sensitivity:

- **CBBA with Partial Release (+LR):** The adapted partial-release baseline equipped with level recourse (+LR) achieves $90.3 \pm 0.9\%$ completion. The difference vs. ACA is +0.63 percentage points (HL median difference +0.00 pp, 95% CI [-0.63, +1.89] pp, Holm-adjusted $p = 1.000$, Bonferroni $p = 1.000$). This demonstrates statistical parity with CBBA-PR+LR across benchmark topologies. In welfare, ACA achieves non-inferiority within the pre-registered margin ($p_{\text{TOST}} = 1.000$).
- **CTP Replan and Implicit Coordination:** In discrete graph simulation, CTP Replan achieves $96.4 \pm 0.5\%$ task completion (-5.5 pp difference vs. ACA, adjusted $p =< 0.001$), and Implicit Coordination achieves $94.5 \pm 0.7\%$ completion (-3.6 pp difference vs. ACA, adjusted $p =< 0.001$). Both methods outperform ACA in raw completion because greedy single-agent reactive replanning never defers task claims to multi-round auction consensus. However, this completion advantage comes at high coordination risk: Implicit Coordination incurs 107.7 ± 4.2 total conflicts and 10.6 ± 0.2

distinct double-assigned tasks, whereas ACA isolates safe claims, achieving 99.1% safe-conflict reduction.

- **Sequential Single-Item (SSI):** In this static task-set benchmark, SSI achieves high completion ($84.3 \pm 1.3\%$) and net welfare (694 ± 28) with zero conflicts and low message volume (252 ± 11). When equipped with recourse (SSI+LR), ACA maintains a significant completion advantage of +6.77 pp ($p < 10^{-5}$).
- **Interdiction Budget Sensitivity (H_6):** Across simultaneous blockage budgets $\kappa \in \{2, 4, 6\}$ ($N = 750$ instances, 3,300 trials; Supplementary Fig. S6), increasing adversary budget monotonically degrades completion across topologies (Jonckheere–Terpstra $p < 10^{-12}$), with ACA maintaining its advantage over CBBA variants.
- **Null Findings on Regimes and Topologies:** Across communication regimes (H_{2_REGIME}), ACA demonstrates parity under lossy conditions ($p = 0.507$). Across topology classes (real vs. synthetic), completion differences are statistically non-significant ($p = 0.840$), demonstrating consistent behavior across map structures (Supplementary Table S7).
- **Clairvoyant Greedy (Oracle Reference):** The centralized omniscient Oracle achieves $98.8 \pm 0.4\%$ completion and 1240 ± 12 welfare with zero conflicts, serving as an informed empirical reference (not a mathematical bound).

Mean per-tick computation time for ACA is 35.52 ± 2.49 ms, operating comfortably within real-time multi-robot planning rates while accounting for contingency tree evaluation and Theory of Mind calculations. Supplementary Table S7 details performance stratified across map classes and adversarial regimes.

D. Protocol v5 Confirmatory Evaluation on Held-Out Seeds

To eliminate hyperparameter tuning leakage and test out-of-sample generalizability across the entire benchmark universe, Protocol v5 was executed across all 5 map topologies and 3 adversary profiles on strictly held-out random seeds $101 \ldots 130$ ($N = 450$ matched independent instances, 3,600 total paired trials). Table III summarizes confirmatory results under family-wide Holm–Bonferroni correction.

TABLE III
PROTOCOL V5 CONFIRMATORY HELD-OUT RESULTS ACROSS 5 BENCHMARK FAMILIES AND 3 ADVERSARY REGIMES ($N = 450$ MATCHED INSTANCES, SEEDS 101–130, MEAN ± 95% BOOTSTRAP CI).

Method	Completion (%)	Net Welfare	Total Conf.	Messages
ACA+ (Ours)	92.4 ± 1.0	936 ± 25	38.6 ± 2.9	221 ± 13
ACA-Adaptive	91.3 ± 1.1	920 ± 25	39.0 ± 3.0	232 ± 14
ACA (Static Pinsker)	92.4 ± 1.1	937 ± 25	39.7 ± 3.0	218 ± 13
ACA (No Margin)	93.5 ± 1.0	939 ± 23	49.9 ± 3.3	213 ± 12
CBBA (+LR) [1]	94.7 ± 0.9	959 ± 22	25.5 ± 1.8	243 ± 11
CBBA-PR (+LR) [11]	93.4 ± 1.0	953 ± 22	40.6 ± 2.8	239 ± 12
CTP Replan [28]	92.3 ± 1.1	540 ± 42	0.0 ± 0.0	56 ± 3
SSI (+LR) [10]	80.5 ± 2.0	666 ± 41	0.0 ± 0.0	146 ± 10

In this held-out evaluation, a headline reversal is observed: CBBA+LR outperforms ACA on task completion (94.7% vs. 92.4%, difference +2.27 pp) and net welfare (958.9 vs. 937.4) with fewer conflicts (25.5 vs. 39.7). This reversal establishes that ACA's advantage is safety coordination via the Pinsker margin under communication constraints rather than unconstrained completion dominance. ACA+ achieves 92.4% completion, demonstrating parity with static ACA (92.4%, difference +0.04 pp, Holm $p = 1.000$). In Protocol v5, pruning was inactive because registered threshold (0.06) fell below candidate admission threshold (0.12); calibrated active pruning ($\tau = 0.15$) is formalized in Protocol v6. Evaluating ACA against ACA (No Margin) reveals the Pinsker margin's central coordination role: ACA reduces total conflicts by 20.6% relative to ACA (No Margin) (39.7 vs. 49.9, difference -10.3, Holm $p =< 0.001$), whereas disabling the margin yields only an unadjusted 1.08 pp completion gain ($p = 0.008$) at high conflict cost. CTP Replan achieves completion parity (92.3%) with fewer messages (56 vs. 221, $p < 10^{-67}$), and ACA+ maintains a +12.0 pp completion gain over SSI+LR ($p < 10^{-25}$).

E. Software-in-the-Loop Simulation (Software Simulation Only)

To evaluate whether algorithmic conclusions hold under continuous physical execution, we conducted Software-in-the-Loop (SITL) simulations using PyBullet rigid-body physics and OpenCV ArUco marker tracking. **All experiments in this section represent pure software simulation; no physical robots were deployed.** Evaluations were performed across $N = 360$ matched paired trials per method over identical random seeds ($20000 \ldots 20359$, disjoint from the primary sweep) spanning all 18 condition cells (2 real maps × 3 adversaries × 3 communication regimes). Table IV summarizes the paired results.

TABLE IV
SOFTWARE-IN-THE-LOOP (SITL) PHYSICS SIMULATION VS. DISCRETE GRAPH SIMULATION ($N = 360$ PAIRED TRIALS PER METHOD ON REAL MAPS).

Method	SITL (%)	Graph (%)	Gap (pp)	p-val	Contacts	Conflicts
ACA (Ours)	69.4 ± 20.3	88.4 ± 16.7	-18.9	–	158.6	37.5
ACA (No Margin Ablation)	68.6 ± 20.4	88.6 ± 14.9	-20.1	0.8358	171.7	66.5
CBBA-PR (+LR)	68.8 ± 19.4	90.3 ± 13.7	-21.5	0.8358	194.7	70.0
CTP Replan	68.4 ± 20.8	91.3 ± 15.7	-22.9	0.8358	142.3	0.0
Implicit (No Consensus) [14]	74.9 ± 19.5	92.1 ± 13.7	-17.2	$< 10^{-4}$	199.8	168.6
SSI (+LR)	42.2 ± 25.4	72.2 ± 30.1	-30.1	$< 10^{-4}$	143.6	0.0

1) Sim-to-Sim Domain Gap and Rank Invariance Failure: Continuous physics introduces velocity limits and spatial proximity interactions absent from discrete simulation. All evaluated methods exhibit a completion drop of ≈ 18.9 pp (ACA: $69.4\% \pm 20.3\%$ in SITL vs. $88.4\% \pm 16.7\%$ in graph simulation; HL median gap -18.8 pp, $[-18.8, -18.8]$ pp, Holm $p =< 0.001$). Rank invariance fails across fidelities: Spearman rank correlation between discrete and continuous rankings is non-significant ($\rho = 0.49, p = 0.33$). Paired tests show parity for CBBA-PR+LR (+0.62 pp, $p = 0.836$), ACA-no-margin (+0.87 pp, $p = 0.836$), and CTP Replan (+1.04 pp, $p = 0.836$), while Implicit achieves higher raw completion (+5.48 pp, $p < 10^{-4}$) with severe collisions (168.6 conflicts/trial vs. ACA's 37.5).

2) Physical Contacts and Vision Performance: Physical collisions correlate with detour activations ($r = 0.775$; Spearman $\rho = 0.747$) as corridor blockages force bottleneck congestion. Synthetic ArUco marker tracking across 20 runs (10/map, 10 seeds, 1000 frames, 4,000 opportunities) yields detection rate 82.9% ($[78.55\%, 86.90\%]$). Mean translational position error was 0.416 m ($[0.257, 0.617]$ m; RMSE 0.688 m). Mean orientation error was 0.0107 rad (0.62°, $[0.0090, 0.0124]$ rad; RMSE 0.0139 rad), satisfying the tracking bound (< 0.10 rad).

VII. Limitations and Future Work

(1) *Simulation Scope:* Evaluations are conducted in discrete graph and continuous PyBullet simulation; hardware multi-robot testbed deployments remain planned future work. (2) *Estimation Error Bounds:* Theorem 1 conditions on $\mathrm{TV}(B_s, \hat{B}_s) \leq \eta_{\text{eff}}$; under severe packet loss, conservative consensus deferral handles bound violations. (3) *Contingency Scalability:* Discrete recourse trees scale with depth d; practical deployments employ depth pruning ($d \in \{1, 2\}$).

VIII. Conclusion

We presented Adversarial Contingency Auctions (ACA) for decentralized multi-robot task allocation under strategic corridor blockages and belief inconsistency. By coupling type-aware expected recourse cost valuation with a closed-form Pinsker claim margin and autonomous local detours, ACA mitigates conflicts without continuous belief synchronization. Across exploratory evaluations, ACA demonstrated coordination gains over CBBA+LR (+2.14 pp, $p = 0.004$) and reduced safe-claim conflicts by 99.1%. However, pre-registered Protocol v5 evaluation on strictly held-out seeds established a performance reversal where CBBA+LR achieved higher completion (94.7% vs. 92.4%, difference +2.27 pp), demonstrating that ACA's primary operational advantage is safety coordination via the Pinsker margin under communication constraints rather than unconstrained completion dominance. Protocol v6 is cryptographically frozen with confirmatory execution pending. Physical multi-robot testbed validation remains future work.

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